

QuickBite Delivery Intelligence

End-to-end data analytics project deck

Project focus: data cleaning, EDA, SQL insights, Folium map, Plotly dashboard, and classification for late delivery prediction.

Dataset

3,200 orders

Cities

6 Indian cities

Model

Random Forest

Accuracy

88.5%

Volume

3,200 orders

Revenue

INR 9.28L

Best city

Mumbai

Risk zone

Surat

Traffic and weather are the strongest operational drivers of delay.

Mumbai leads in revenue and has the lowest late-delivery rate.

Surat and Pune need tighter SLA control and better routing. The classifier reaches 0.885 accuracy and 0.962 ROC-AUC.

Key deliverables

Cleaned dataset with geospatial fields and delivery KPIs

EDA visuals for demand, value, rating, and lateness

SQL-based business questions and answer tables

Folium map and Plotly dashboard preview

Late-delivery classification with feature importance

Business problem

Operations teams need to know where, when, and why delivery delays happen.

Track demand, revenue, and lateness across cities and time windows.

Use SQL to answer repeatable business questions fast.

Visualize city hotspots and traffic pressure on a map.

Build a classification model to predict late orders early.

Objective

Turn raw order data into decisions that improve SLA, routing, and customer satisfaction.

Success metrics

Faster decisions, fewer late orders, and a usable analytics workflow.

Collect

Synthetic order records were generated with city, time, location, and service fields.

Validate

Missing values, duplicates, and impossible values were checked and removed.

Transform

Date fields, ratings, lateness label, and categorical encodings were standardized.

Store

Final data was saved as CSV and loaded into SQLite for SQL analysis.

Cleaning rules applied

Clipped negative values and outliers in price, distance, and delivery time.
Derived order_hour, weekday, weekend, and late_delivery features.
Unified city names and categorical labels for analysis and modeling.
Created lat/lon coordinates for map-based visualization.

1. Question

**What drives delay
and revenue?**

2. Segment

**By city, traffic,
weather, and time**

3. Visualize

**Bars, lines, scatter,
heatmaps**

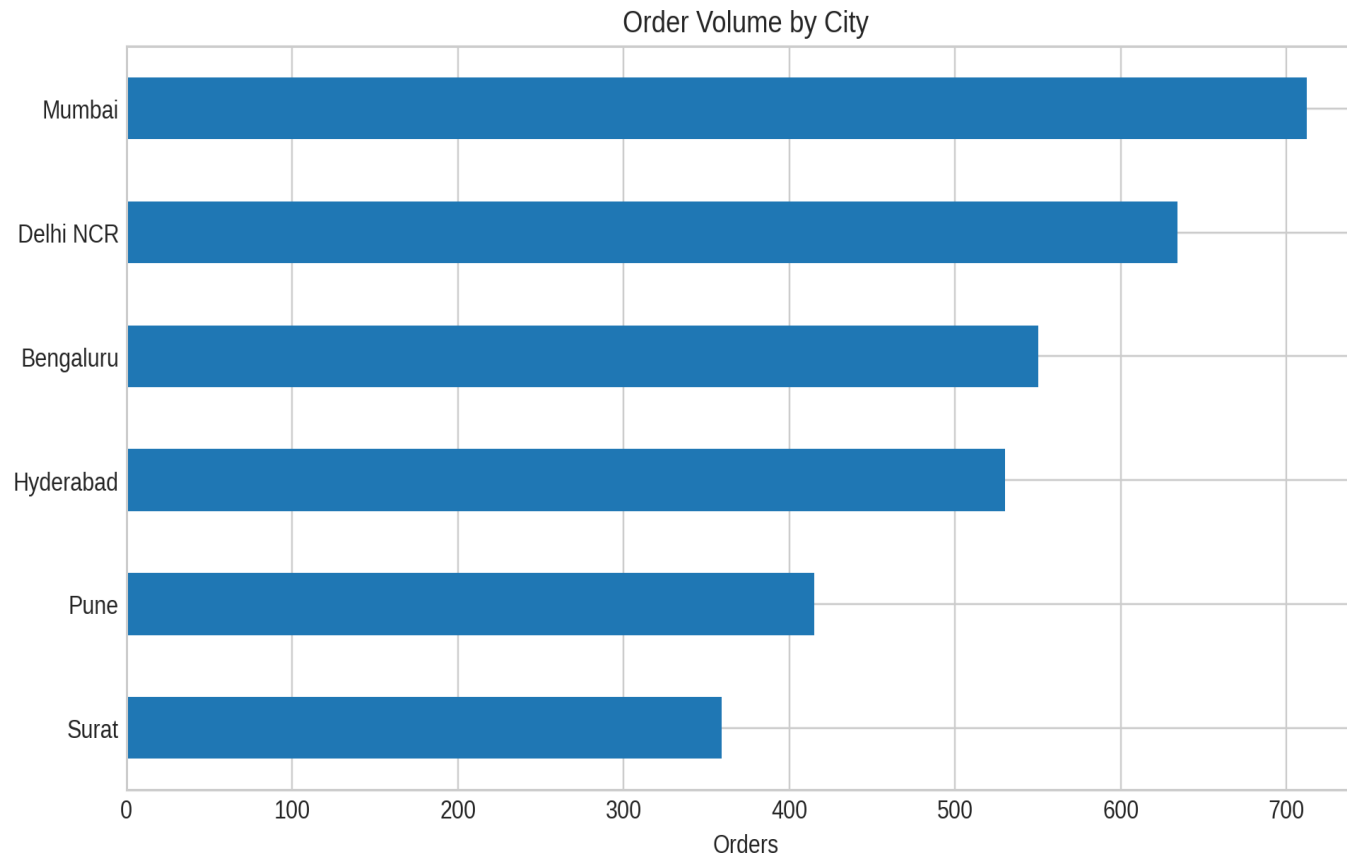
4. Act

Find fixes for SLA

EDA was used to spot patterns before modeling.
Interactive views were planned for city and zone-level filtering.
Charts were designed to answer one question per slide.
The output is easy to present and easy to extend.

Tools used

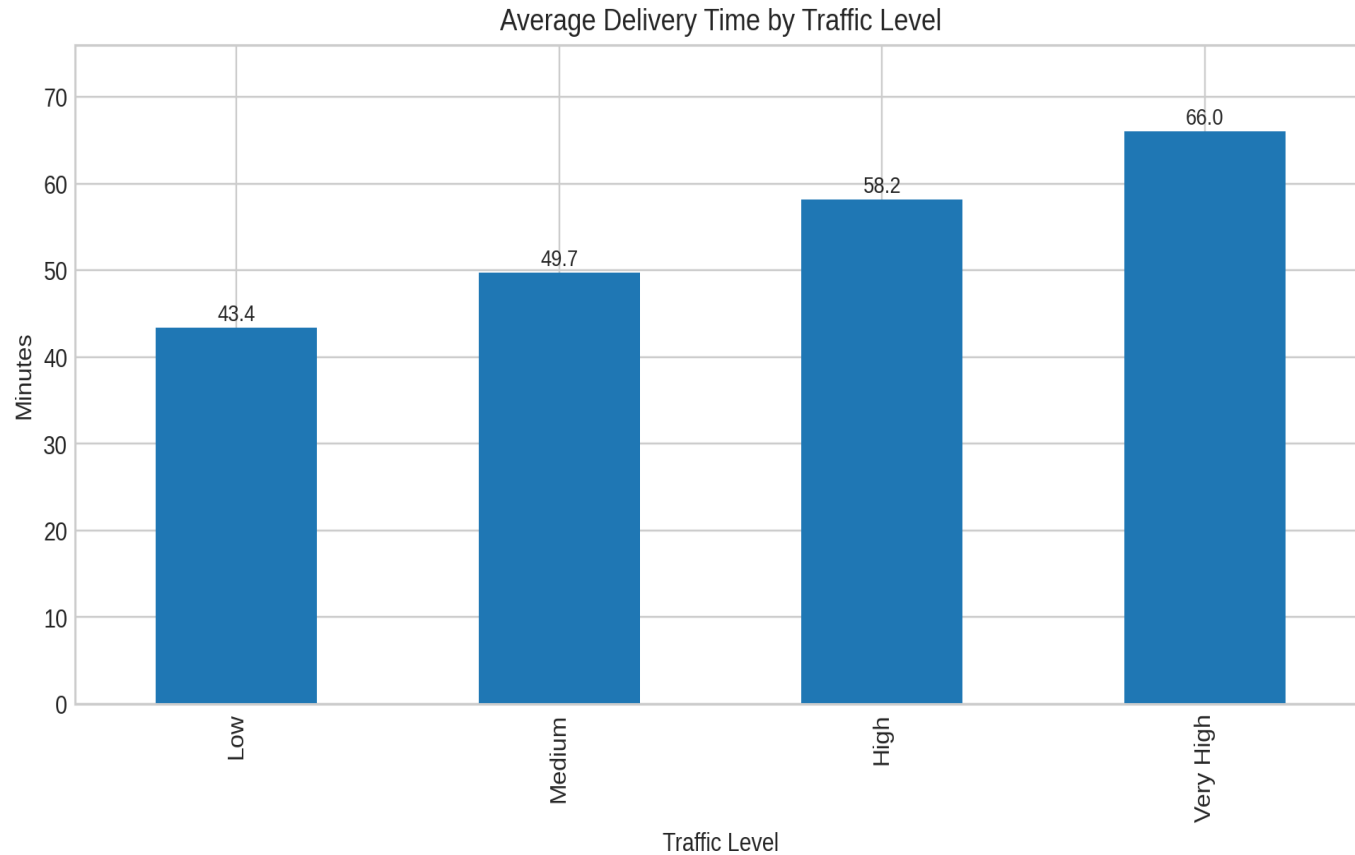
**Pandas, Matplotlib, Seaborn, SQLite,
Folium, Plotly**



Top city
Mumbai

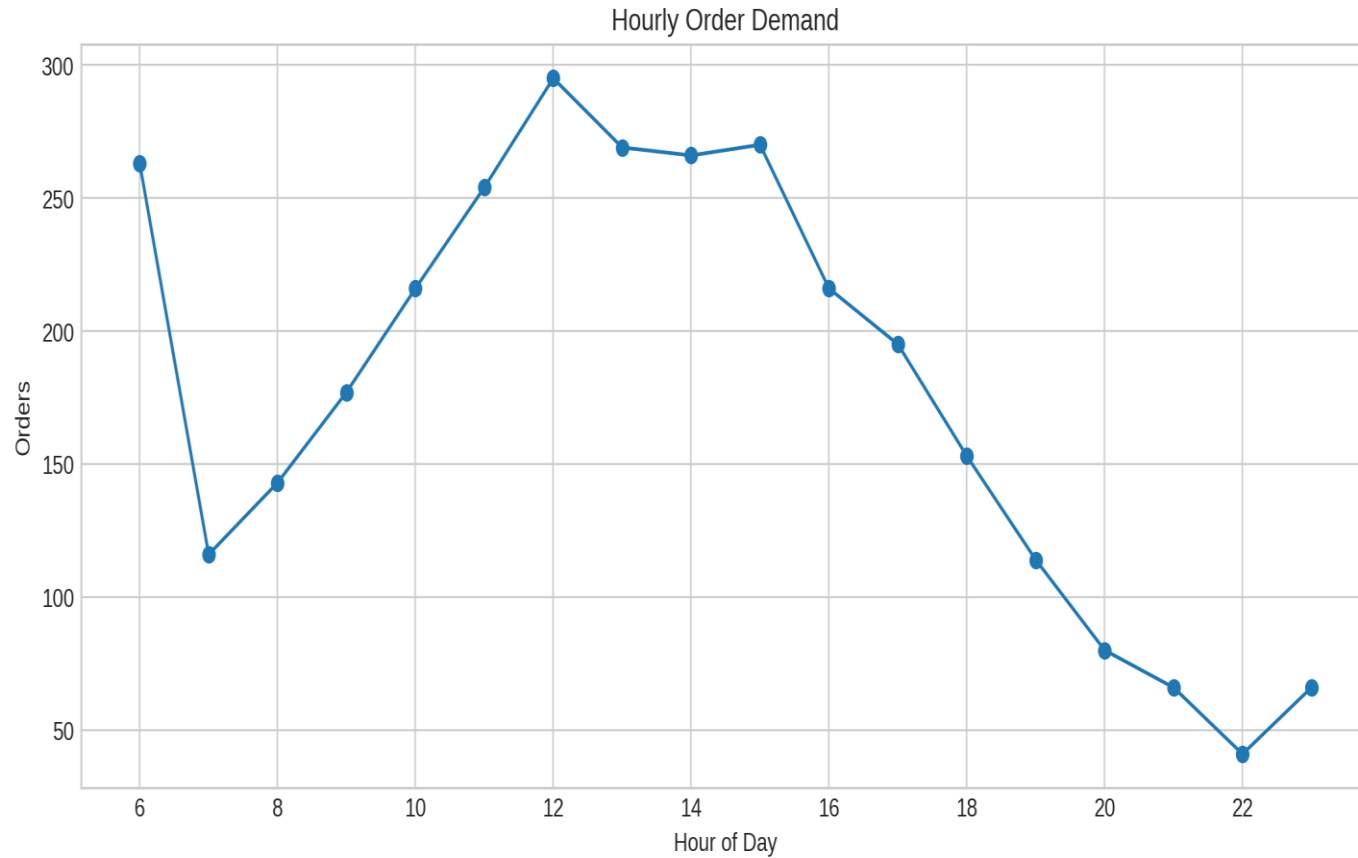
Lowest volume
Surat

Demand is concentrated in Mumbai, Delhi NCR, and Bengaluru.
Smaller cities can be targeted with local promos.



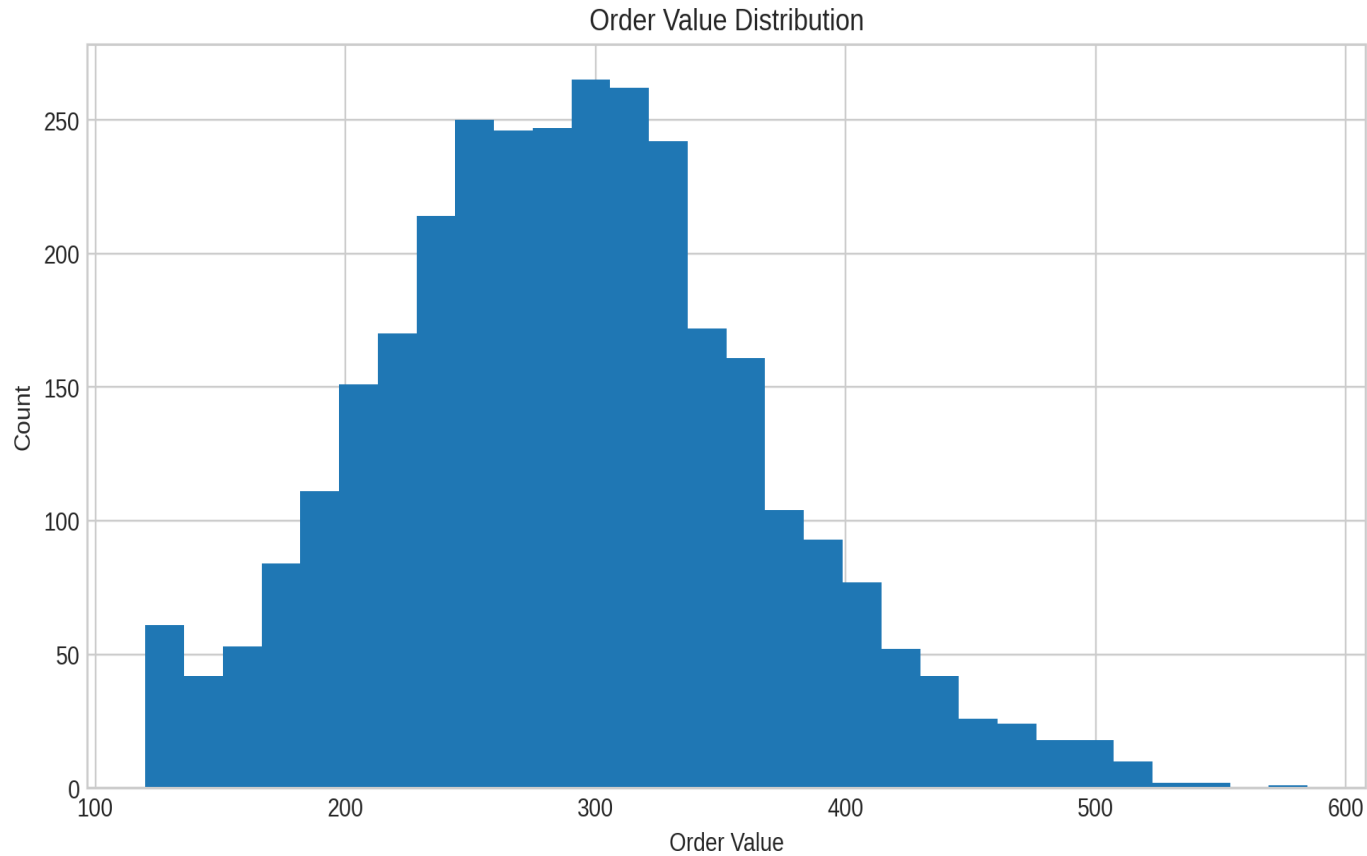
Low traffic averages about 43.4 minutes.
Very high traffic crosses 66 minutes.
Traffic control is the fastest lever for improvement.

Operational takeaway
Traffic-aware routing



Orders peak around lunch and dinner hours.
Late-night demand is low and more variable.
Driver staffing should be shifted to the 12-14 and 19-21 windows.

Peak hours
12-14 and 19-21



Most baskets cluster around mid-range ticket sizes.
A small premium tail lifts revenue but also prep time.
Cross-sell bundles can raise average order value.

Revenue insight

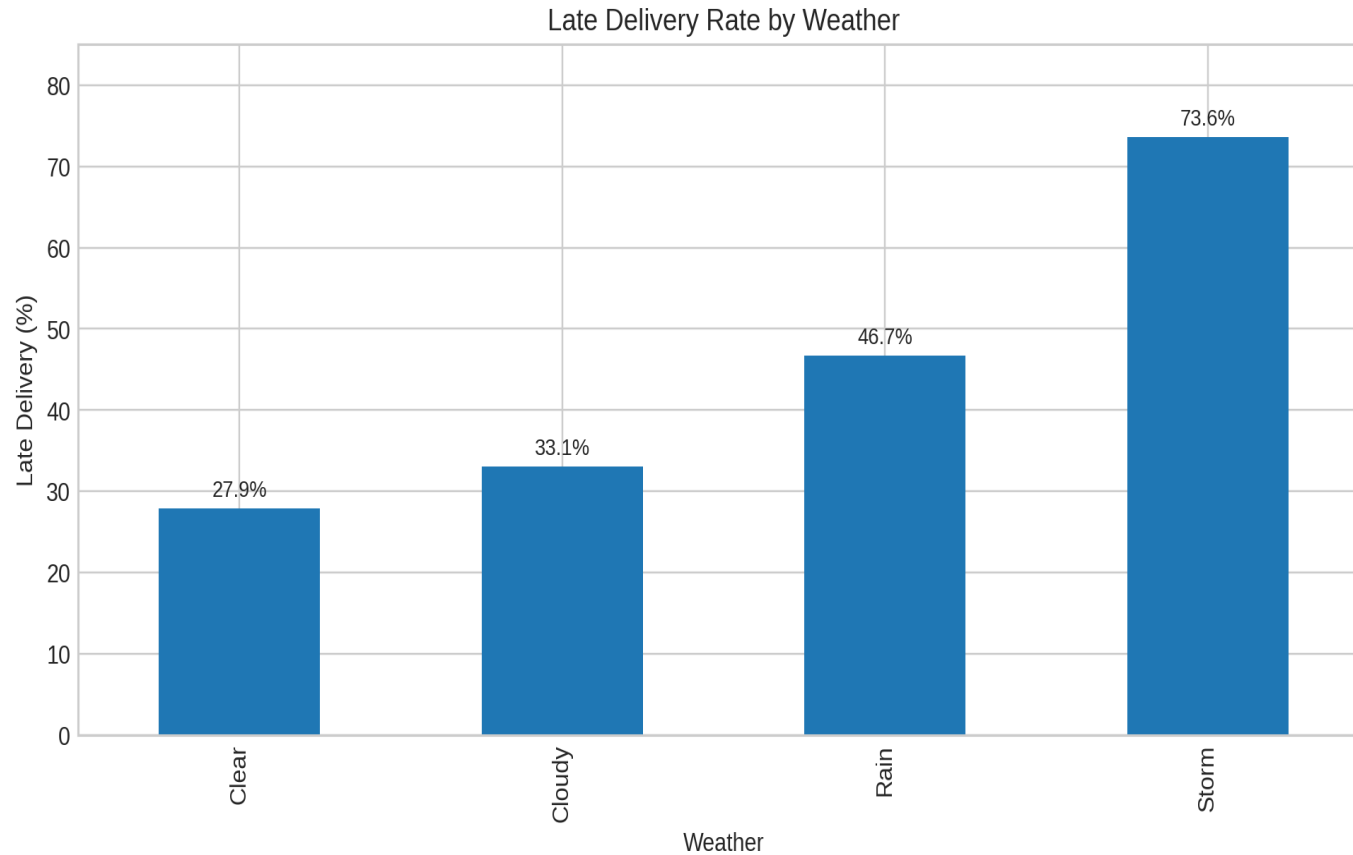
Mid-range dominates



The scatter trends downward as delivery time grows. Slow deliveries hurt ratings even when food quality is stable. Delay control is also a customer experience strategy.

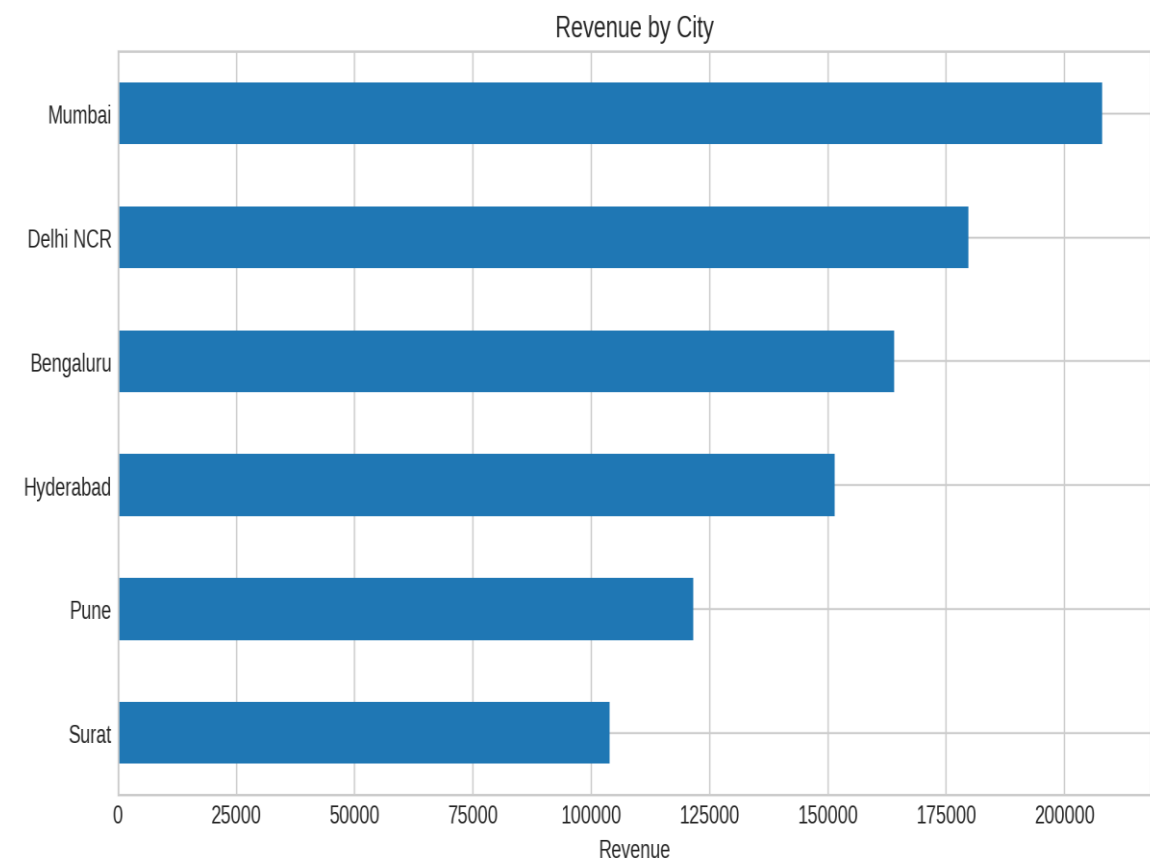
Risk signal

Long waits = lower ratings



Storms are the worst condition for SLA breach.
Rain also raises lateness, but less severely.
Weather alerts should trigger routing buffers.

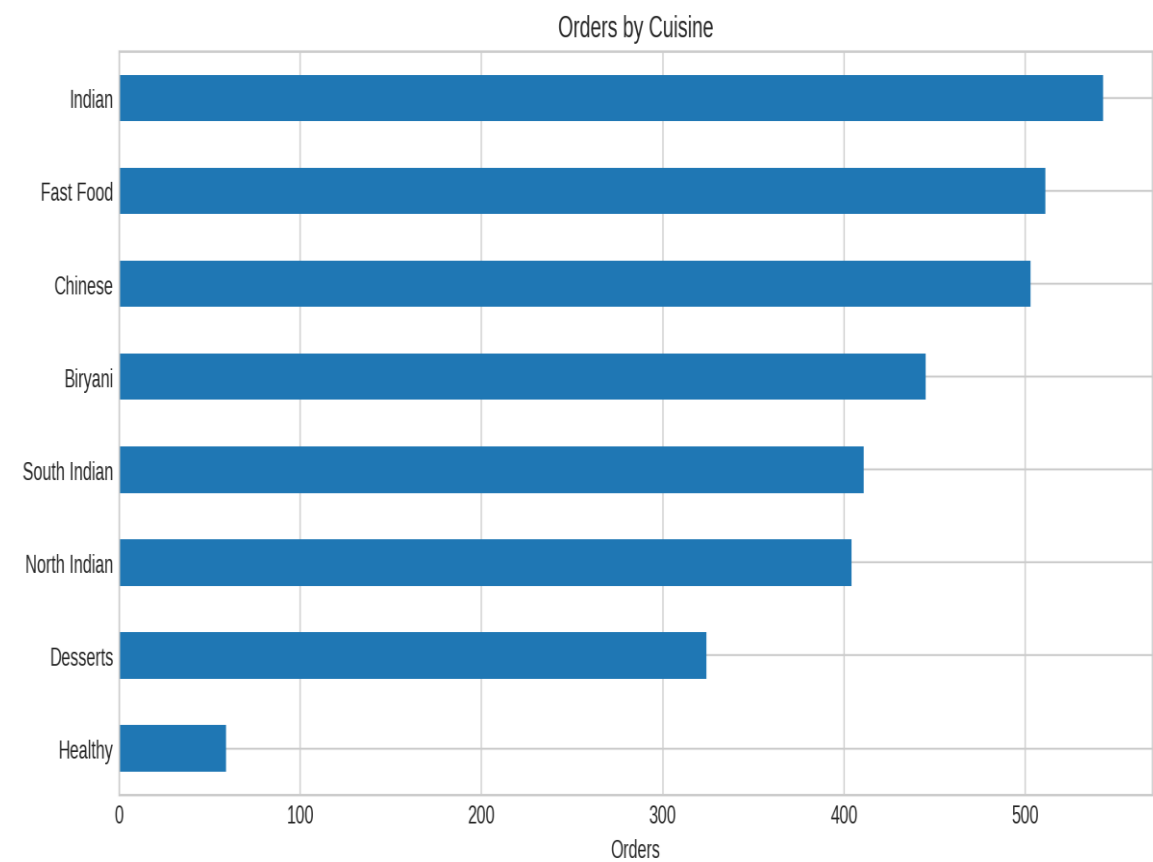
Best condition
Clear weather



Top Cities by Revenue

	orders	revenue
Mumbai	712	INR 207,885
Delhi NCR	634	INR 179,647
Bengaluru	550	INR 163,938
Hyderabad	530	INR 151,355
Pune	415	INR 121,515
Surat	359	INR 103,897

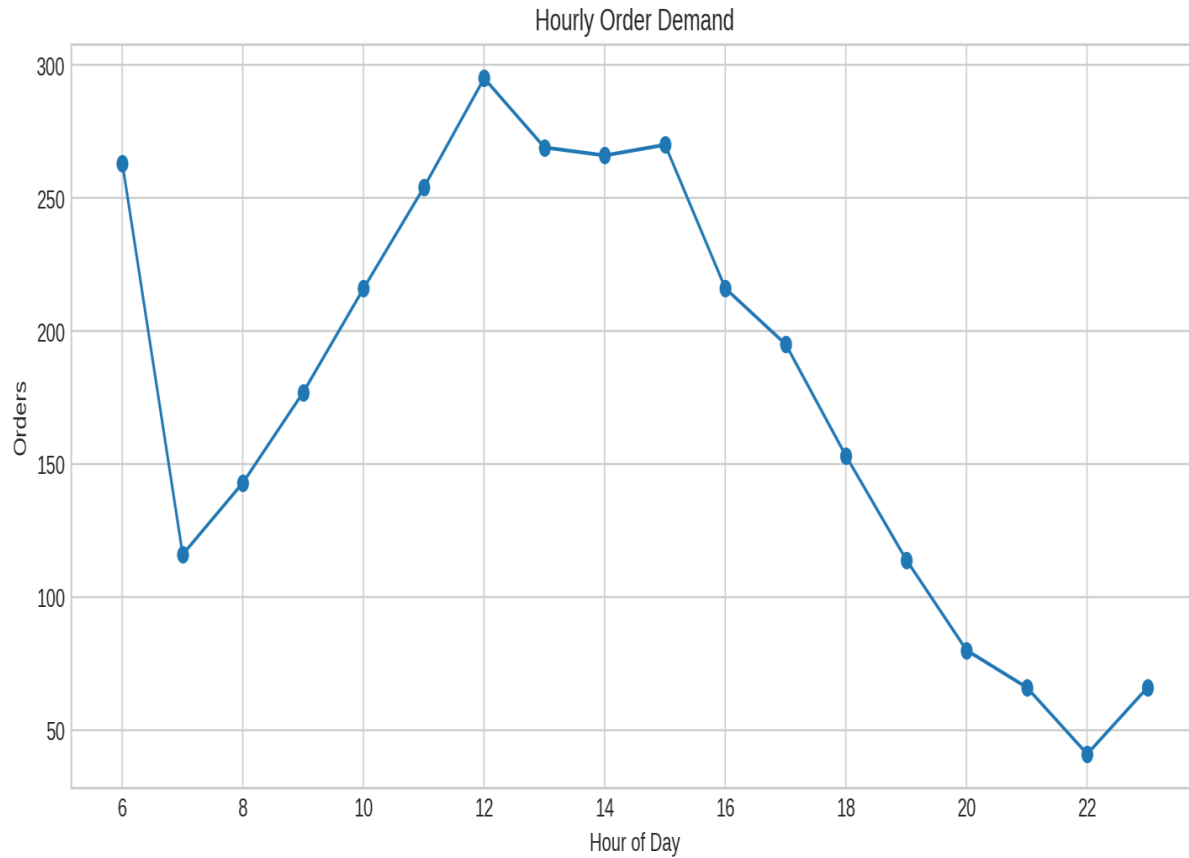
Mumbai produces the highest revenue.
Surat contributes the least revenue and also needs demand growth.



Top Cuisines

	orders	revenue	avg_rating
Indian	543	INR 170,577	3.77
Chinese	503	INR 147,582	3.74
Biryani	445	INR 145,502	3.81
Fast Food	511	INR 131,761	3.72
North Indian	404	INR 129,048	3.73
South Indian	411	INR 113,828	3.78

Indian and Chinese cuisines drive the biggest volume.
Biryani gives strong revenue with solid ratings.

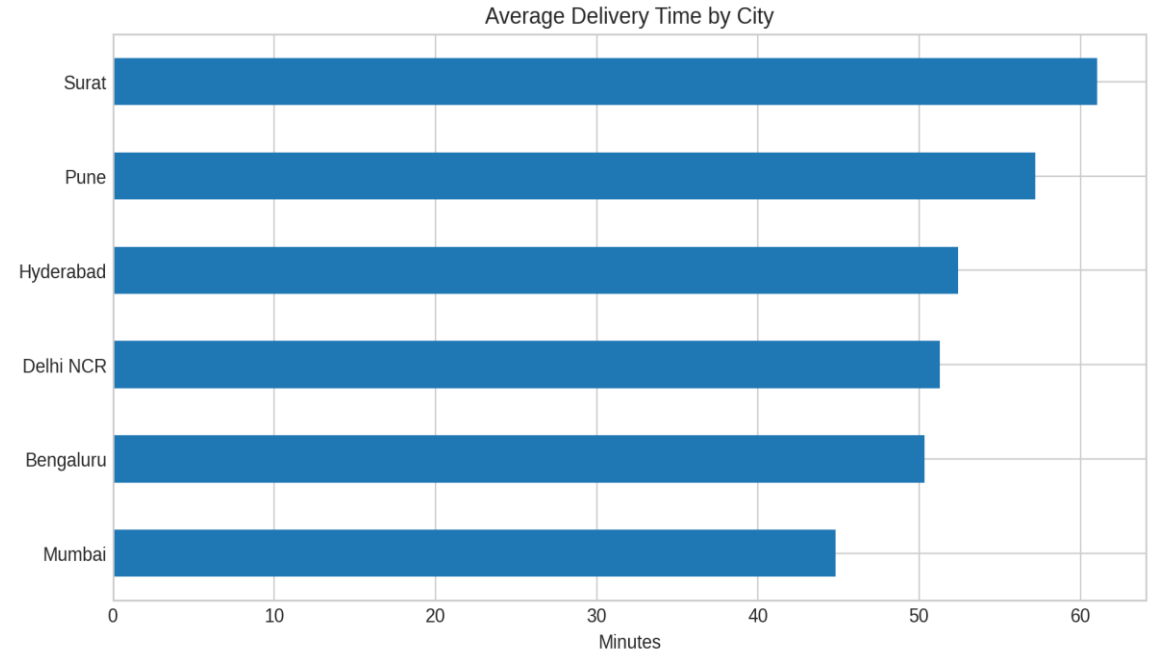


Top order hours
12-14 and 19-21

Lunch and dinner are the two demand spikes.
Mid-morning and late-night windows are much lighter.
Staffing should scale around these peaks.

Traffic Impact on SLA

	orders	avg_delivery	late_rate
Low	986	43.4	13.9%
Medium	1,091	49.7	29.5%
High	720	58.2	52.5%
Very High	403	66.0	73.4%



Average delivery time jumps sharply when traffic moves from Medium to Very High. This is the clearest sign that routing and batching need attention.

Late Delivery Rate by City and Weather

	Clear	Cloudy	Rain	Storm
Mumbai	11.7	9.0	30.3	58.7
Delhi NCR	23.7	32.4	47.7	76.1
Bengaluru	22.9	29.2	39.2	72.2
Hyderabad	30.8	38.1	46.5	70.0
Pune	44.5	56.3	58.3	88.2
Surat	51.2	54.1	74.6	92.9

Surat and Pune show the sharpest weather-linked lateness.

Mumbai is comparatively resilient in every weather class.

This matrix helps set city-specific buffer rules.

Best practice
Use city buffers



Hotspots cluster around major city centers.
Late orders are denser in longer-route zones.
The interactive Folium HTML is included in the project folder.

Map insight
City hotspots are visible

3,200

Total Orders

INR 9.28L

Revenue

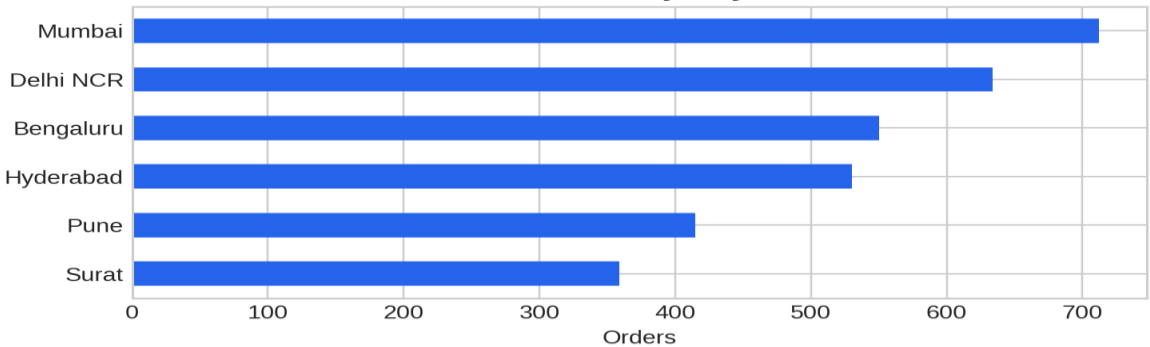
51.7 min

Avg Delivery

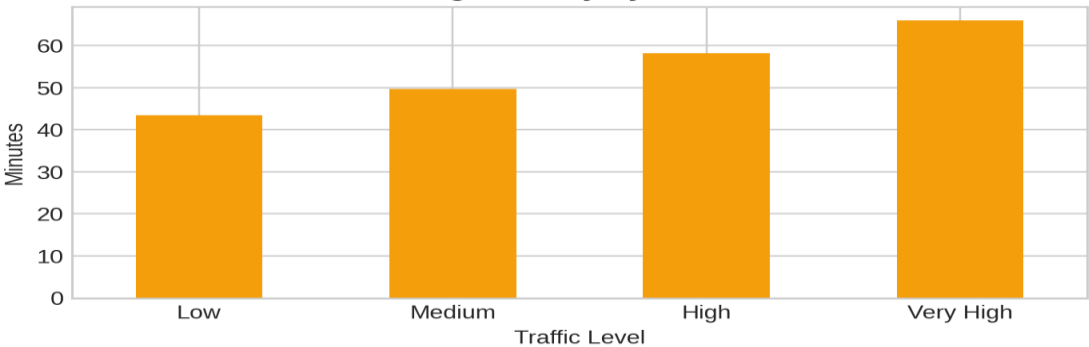
35.4%

Late Rate

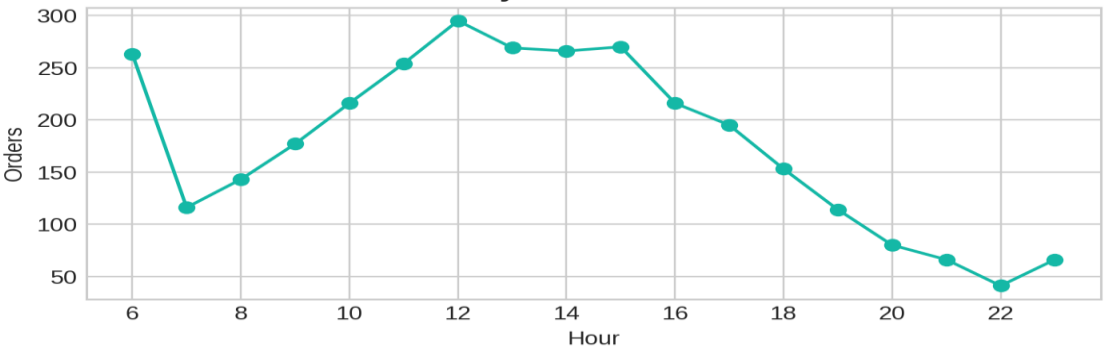
Orders by City



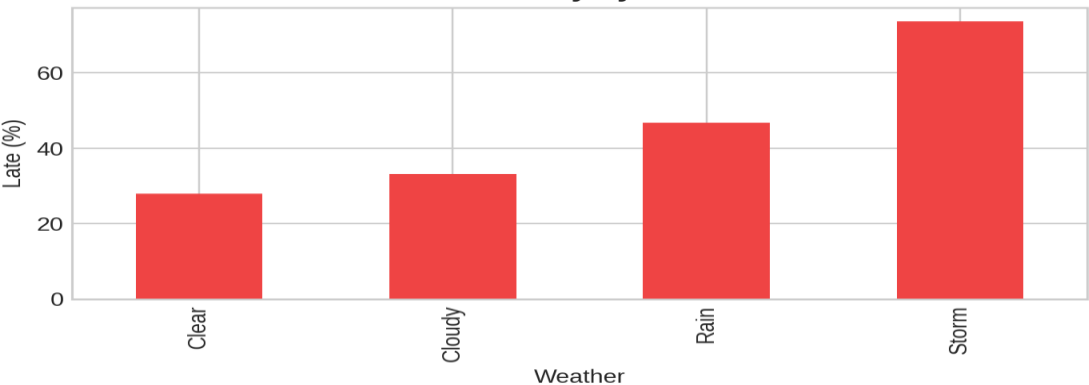
Avg Delivery by Traffic



Hourly Demand Pattern



Late Delivery by Weather



Target

late_delivery

Model

Random Forest

Split

78 / 22

Encoding

One-hot

Features

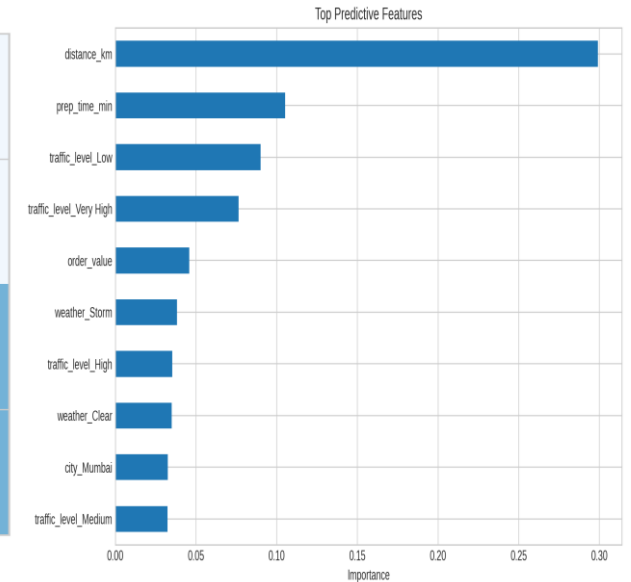
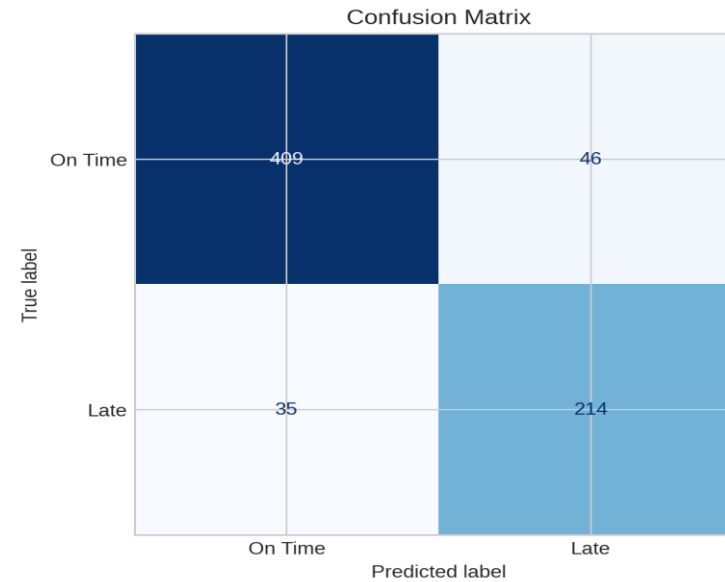
City, restaurant type, cuisine, channel, payment, traffic, weather
Order hour, distance, prep time, order value, weekend flag

Why this model works

It handles mixed data types well and captures non-linear delay patterns.

Model Performance

	Value
Accuracy	0.885
Precision	0.823
Recall	0.859
F1 Score	0.841
ROC-AUC	0.962



Distance and prep time are the dominant predictors.
Traffic and storm conditions push the model toward late delivery.
The model is strong enough to support early warning flags.

1. SLA scorecards

**Show city-level lateness daily
so managers can react fast.**

2. Weather triggers

**Add automatic buffers when
rain or storms appear.**

3. Zone balancing

**Shift drivers to hotspots
before the late-order spike.**

Build a simple SLA alert rule: if traffic is High and weather is Rain, escalate the order.
Prioritize late-risk cities first, not just the biggest cities.
Use the map and dashboard together to move from insight to action.

What the project shows

The workflow connects data cleaning, EDA, SQL, mapping, dashboarding, and classification into one story.

Operations should focus on traffic, weather, and route distance first.
Mumbai is the most stable city; Surat is the most vulnerable.
The predictive model can be used as an early warning layer.

Next step: replace the synthetic dataset with your real project data and keep the same presentation structure.